



Chandra | AI Educator ✓

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**AI Engineer questions interviewer
expects you to know:**



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Stats & Math for ML:

- What is the mean, and how is it used in model evaluation or dataset analysis?
- When is the median a better measure than the mean in real-world ML datasets?
- What does variance represent in a dataset used for training ML models?
- How is probability formally defined, and why is it fundamental to AI models?
- What is conditional probability, and how is it used in classification models?
- What is Bayes' Theorem, and how would you explain it intuitively in an interview?



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Machine Learning Fundamentals:

- What is Machine Learning, and how is it different from traditional programming?
- What is the difference between AI, ML, and Deep Learning?
- What is supervised learning?
- How do regression and classification differ in output and loss functions?
- Why do we split data into training and testing sets?
- What is overfitting in Machine Learning?



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Deep Learning Basics:

- Why was deep learning needed when traditional machine learning already existed?
- What is an artificial neuron, and how does it relate to a biological neuron?
- What happens during the forward pass in a neural network?
- What is backpropagation, explained intuitively without math?
- Why are activation functions necessary in neural networks?
- Why is ReLU commonly preferred in deep networks?



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LangChain:

- What is LangChain, and how is it used in NLP applications?
- What are the core components of LangChain?
- What is a chain in LangChain, and how is it used in NLP?
- What is a LangChain agent, and how is it used in NLP?
- What is the ReAct framework in LangChain agents?
- What is memory in LangChain, and how is it used in NLP?



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RAG Interview:

- What is Retrieval Augmented Generation (RAG), and why is it critical for LLMs?
- What are the two main components of a RAG application architecture?
- What is the step-by-step workflow of a complete RAG pipeline?
- Why is it necessary to split documents into smaller chunks?
- What is the purpose of embedding text and storing it in a vector store?
- What is the specific role of a Retriever in LangChain?



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LangGraph:

- What is LangGraph, and when would you choose it over LangChain Chains or Agents?
- What makes LangGraph "stateful," and why is state central to its design?
- What are nodes and edges in LangGraph, and what do they represent in execution?
- What is the Graph API in LangGraph, and how do you build a basic graph end-to-end?
- How do you design a state schema for a production agent (minimal vs rich state)?
- How do conditional edges work in LangGraph, and when do you use them?



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CrewAI:

- What is CrewAI and what problem does it solve in agentic AI systems?
- How is CrewAI different from LangChain Agents and LangGraph?
- What are the core components of CrewAI (Agent, Task, Crew, Process)?
- What is an Agent in CrewAI, and what responsibilities does it have?
- What is a Task in CrewAI, and how does it differ from an agent?
- What is a Crew in CrewAI, and how does it manage execution?



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AutoGen:

- What is Microsoft AutoGen, and what problems does it solve compared to single-agent chatbots?
- What's the difference between an AssistantAgent, UserProxyAgent, and tool-enabled agent in AutoGen?
- How do you design roles and system prompts for multiple agents so they don't overlap or argue endlessly?
- How do you register tools/functions in AutoGen and control what each agent is allowed to call?



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Agentic AI:

- What do you understand about Agentic AI?
How is it different from Traditional AI?
- What are the key components of an Agentic AI system?
- Explain the end-to-end process of building an AI agent, from ingestion to evaluation.
- What is tool use and function calling in LLMs, and why is it essential for agents?
- What are autonomous agents vs. assisted agents, and when is each preferred?
- How does memory work in AI agents (short-term, long-term, episodic)?



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FastAPI:

- What is FastAPI?
- Explain the asynchronous features of FastAPI.
- Can you explain the key differences between FastAPI and Flask and why you would choose FastAPI over Flask?
- What is the role of Pydantic in FastAPI?
- How does FastAPI leverage type hints for automatic API documentation?
- Can you explain how you would implement authentication and authorization in FastAPI?



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